

Questions

Q1.

Supermarket price war, food waste and possible mergers

Figure 1: Market shares by total revenue in the UK supermarket sector, 2010 and 2015

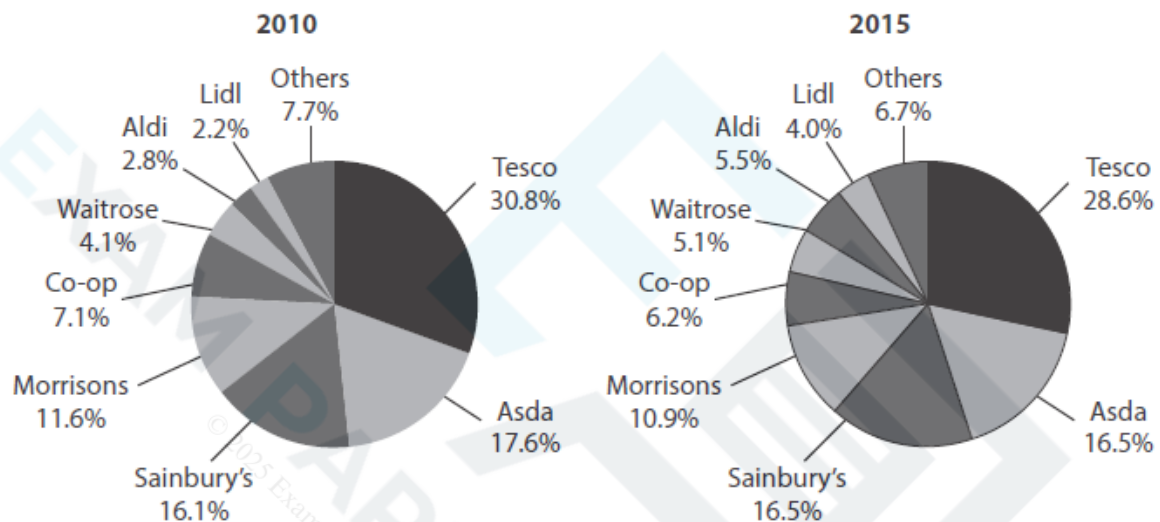
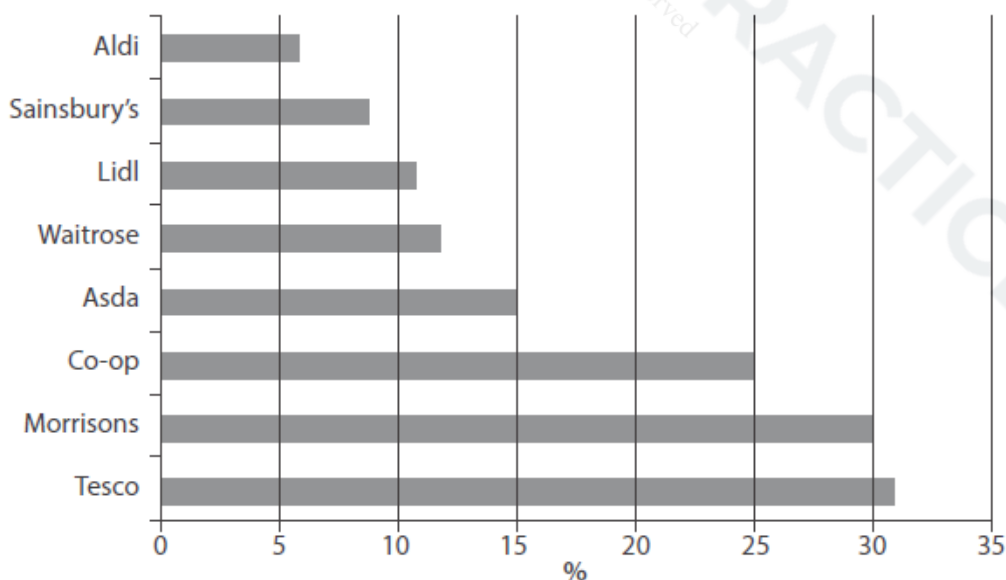


Figure 2: Proportion of food suppliers reporting that the following supermarkets fail to meet the Groceries Code of Conduct, 2015.



Extract A

Supermarket price war puts pressure on their food suppliers

The number of food suppliers (to supermarkets) struggling to remain in business has increased by more than 50% over the past year as supermarkets engage in an intense price war. It has never been tougher for the UK's food suppliers according to a study by accountants Begbies Traynor. It blames aggressive price-cutting by the supermarkets and delays in payments to food suppliers as the main causes of the difficulties. Further problems include food suppliers being forced to pay excessive amounts for packaging specified by supermarkets and funding in-store promotions. Almost 90% of struggling food suppliers are small and medium-sized businesses. 5

The price war has contributed to food prices paid by consumers falling by 1.7% over the past two years. 10

The market shares of the big four supermarkets – Tesco, Asda, Sainsbury's and Morrisons – are under pressure as shopping habits change. Many consumers are switching from one main weekly shop to shopping more frequently at local discount stores such as Aldi and Lidl or purchasing goods online from other grocery retailers. 15

The big four supermarkets have responded by putting more pressure on their suppliers despite an investigation by the Groceries Code Adjudicator (GCA). The GCA has the power to fine supermarkets up to 1% of their annual sales revenue if they break the Groceries Code of Conduct. A YouGov study found considerable differences between the supermarkets in meeting the Code with Aldi performing well but Tesco badly. Despite the Groceries Code, many food suppliers are reluctant to complain for fear of losing contracts with the supermarkets. 20

Extract B

Food waste in the supply chain

A report from the British Retail Consortium reveals that supermarkets are directly responsible for around 0.2 million tonnes of food waste every year. This is due to the expiry of use-by-dates and poor handling of stock.

However, 4.1 million tonnes of food waste occurs annually in the food supply chain before it even reaches the supermarkets, indicating the existence of information gaps. The supermarkets are cooperating with food suppliers and farmers to try to reduce this waste. This involves improving forecasts for supply and demand of food and increasing the reliability of transportation and storage. 5

Consumers, the final stage of the supply chain, waste a further 7 million tonnes of food each year. This suggests irrational behaviour. Supermarkets are also working with consumers to reduce the waste by providing advice on how to store and use leftover food. The development of packaging designs to keep food fresher for longer is one of the innovations under way to reduce waste. 10

Extract C

Proposed merger activity in the supermarket sector

Analysts at Société Générale, an investment bank, have recommended a merger between Sainsbury's and Morrisons. They claim it would lead to increased economies of scale and market power for the combined business. Such a merger between the third and fourth largest supermarkets in Britain would have been unrealistic a few years ago due to concerns of its impact in reducing competition. However, the chances of getting permission from the Competition and Markets Authority have increased following the growth of Aldi and Lidl. Giant mergers have been approved in other sectors such as Lloyds-HBOS (banking) and British Telecom-EE (telecommunications).

The suggested merger would have its challenges. There is considerable overlap between the locations of the stores and the enlarged company would require the rationalisation and co-ordination of hundreds of thousands of employees. A new expensive IT system is likely to be required and the underlying difficult market trends would remain in the food retailing industry.

and Extract A, explain **one** likely reason for the change in the four-firm concentration ratio of the supermarket sector between 2010 and 2015.

(5)

(b) With reference to Figure 2 and Extract A, discuss the possible impact of supermarket monopsony power on both food suppliers and consumers.

(12)

(c) Examine measures the government might use to restrict the monopsony power of supermarkets.

(8)

(d) Assess the extent to which 'information gaps' (Extract B, lines 5 and 6) and 'irrational behaviour' (Extract B, line 11) are the main causes of food waste in the UK.

(10)

(e) Discuss the likely problems for Sainsbury's and Morrisons if the suggested merger between them goes ahead. Refer to Figure 1, Extract C and your own knowledge in your answer.

(15)

(Total for question = 50 marks)

Q2.

The table shows market data for e-cigarette kits. The original equilibrium price is £23.

Price £	Quantity demanded per month (000)	Quantity supplied per month (000)	New quantity demanded per month (000)	New quantity supplied per month (000)
25	5	9		
24	6	8		
23	7	7		
22	8	6		
21	9	5		

As a result of a successful advertising campaign, demand increased by 3 000 e-cigarette kits at all prices. At the same time production costs fell leading to an increase in supply of 1 000 e-cigarette kits at all prices.

(a) Calculate the new equilibrium price and quantity following the successful advertising campaign and the fall in production costs. Use the last two columns for your working.

(4)

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(b) Research conducted in New Zealand in 2014 estimated the cross elasticity of demand for e-cigarettes to be 0.16 in response to changes in the price of tobacco. This implies that a 5% increase in the price of tobacco will cause the percentage change in demand for e-cigarettes to be:

(1)

- ☐ A -31.25
- ☐ B -0.16
- ☐ C 0.8
- ☐ D 31.25

(Total for question = 5 marks)

Q3.

Between 2010 and 2015 the average price of tea in the UK increased from £7.20 per kilo to £8.48 per kilo. Over the same period the quantity of tea purchased fell from 97 million kilos to 76 million kilos.

(Source: <http://www.theguardian.com/lifeandstyle/shortcuts/2015/aug/05/brew-do-you-think-you-are-why-tea-needs-to-copy-coffee-in-order-to-survive>)

(a) Assume that the change in the quantity of tea purchased was only caused by the change in the price of tea. Calculate the price elasticity of demand for tea in response to the rise in its price. You are advised to show your working.

(2)

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(b) Calculate the change in total sales revenue for UK tea retailers between 2010 and 2015. You are advised to show your working.

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(c) Estimates for the demand for black tea in the UK suggest that it is an inferior good. This implies it has a negative:

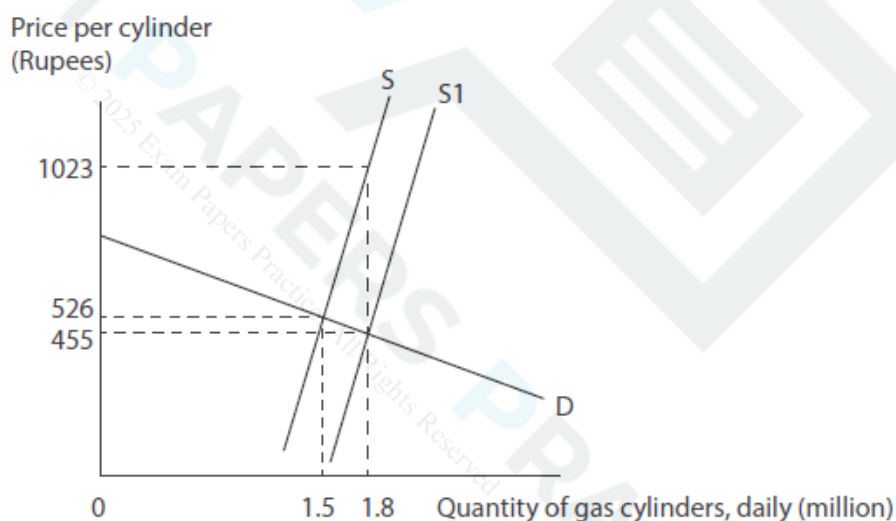
(1)

- ☐ A cross elasticity of demand
- ☐ B income elasticity of demand
- ☐ C price elasticity of demand
- ☐ D price elasticity of supply

(Total for question = 5 marks)

Q4.

The Indian government has introduced a subsidy on domestic cooking gas cylinders in the city of Mumbai. The diagram below shows the effect following the introduction of the subsidy.



(a) Calculate the total domestic cooking gas subsidy paid daily by the Indian government. You are advised to show your working.

(2)

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(b) Calculate the percentage share of the total subsidy received by:

(i) consumers

(1)

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(ii) producers

(1)

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(c) One effect of the subsidy is to increase:

(1)

- ☐ **A** consumer surplus
- ☐ **B** market failure
- ☐ **C** public good provision
- ☐ **D** tax revenue

(Total for question = 5 marks)

Q5.

The following table shows the price and annual production of bags of Arabica coffee beans.

Year	Price per bag on 31 October	Quantity of coffee bean bags (millions)
2013	\$2.84	145
2014	\$4.98	145.2

(Sources: http://ycharts.com/indicators/world_coffee_arabica_price
<http://icocoffeeorg.tumblr.com/post/99916537525/coffee-season-2013-14-finishes-in-balance-but-deficit>)

(a) Calculate the price elasticity of supply of Arabica coffee bean bags between 2013 and 2014.

(2)

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(b) Explain what is likely to happen to the price elasticity of supply for coffee in the long run.

(2)

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(c) The price increase between 2013 and 2014 will:

(1)

- A** signal to producers that coffee has become less popular
- B** ration the good so that more can now afford to purchase it
- C** lead to producers substituting to alternative crop
- D** create an incentive for producers to grow more coffee

Answer

(Total for question = 5 marks)

Q6.

Read Figures 1 and 2 and extracts A and B before answering this question.

UK branded coffee shop chains market

Figure 1: Branded coffee shops by number of shops in the UK in 2021

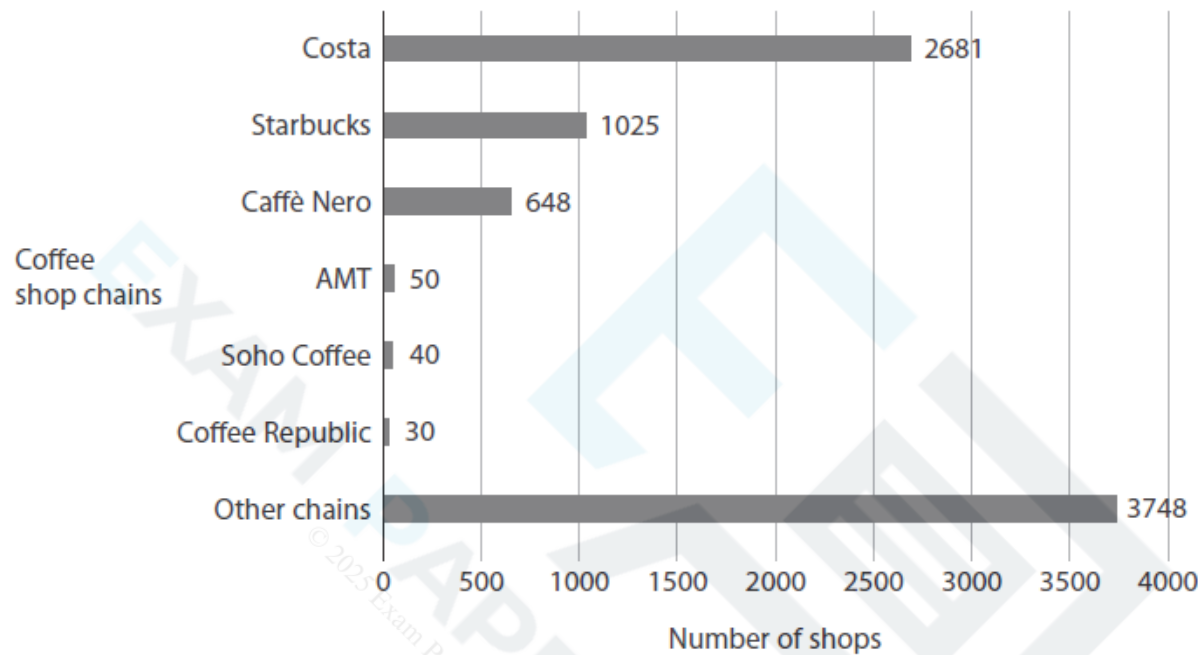


Figure 2: International coffee price (Arabica bean), US dollars per kilogram

Month	Price	Change
Jan 2020	3.13	–
Feb 2020	2.99	–4.47%
Mar 2020	3.27	9.36%
Apr 2020	3.41	4.28%
May 2020	3.30	–3.23%
Jun 2020	3.12	–5.45%
Jul 2020	3.24	3.85%
Aug 2020	3.60	11.11%
Sep 2020	3.67	1.94%
Oct 2020	3.35	–8.72%
Nov 2020	3.31	–1.19%
Dec 2020	3.48	5.14%

Extract A

Tough market conditions for coffee shops – but coffee quality is king

UK coffee shop chains have experienced slow growth opportunities and rising costs. In 2019 the UK market leader, Costa Coffee, opened over 60% fewer stores than in 2018, while Starbucks opened just three new stores overall in 2019.

By 2020, many costs were rising: staff shortages meant rising wages for baristas (trained coffeemakers), a 6.2% National Minimum Wage increase for over 25-year-olds and rising rents. 5

In a challenging UK economy, consumers placed coffee quality ahead of convenient location when choosing a coffee shop. This demonstrates the need for coffee shops to match rising expectations in the UK's increasingly crowded coffee shop market in order to stay competitive. Independent coffee shops (total 25 892 shops in 2020) remain a threat to the branded coffee shops as they pursue a unique luxury experience for customers. This will often focus on the atmosphere and customer service, luxury food and drink ranges and being a part of the local community. Independent coffee shops run on average profit as low as 2% of revenue, and many go out of business as new chains arrive in a locality. 10 15

In January 2020, Coca-Cola finalised its £3.9 billion takeover of market leader, Costa Coffee. The Coca-Cola company's stated aims are to maximise long-term returns to shareholders while being mindful of overall responsibilities such as supporting sustainable communities. Major brands, such as Costa, continue to lead coffee shop expansion in 2020 as competition intensifies. 20

Extract B

Greggs aiming for 'coffee shop experience'

Greggs is looking to increase its branded coffee shop market share after the British bakery chain achieved record profits and overtook Starbucks in the amount of coffee it served in the UK. Although Greggs will continue to focus on the food-to-go market, it will be investing in better coffee machines and providing WiFi. It wants one in three new outlets to have more seating. The company believes that it has the ability to take market share in two ways, by expanding its coffee range and at the same time enhancing the coffee shop experience. Greggs is modelling its plan on rival chain Pret A Manger, which has changed from focusing on food-to-go sales to mainly eating in. Greggs has transformed itself from an unremarkable seller of bread and pastries into one of the UK's most popular food-on-the-go chains, in part thanks to its embrace of vegan, gluten-free and healthy options as well as clever marketing that have changed its image. 5 10

(a) With reference to Figure 1, calculate the three-firm concentration ratio for branded coffee shop chains.

(5)

(b) With reference to Figure 2 and your understanding of price elasticity, examine **two** factors that may cause significant changes in the international price of coffee beans.

(8)

(c) Profit maximisation is assumed to be the business objective of most firms. With reference to Extract A, assess whether this is the case for coffee shop owners.

(10)

(d) Using a cost and revenue diagram, discuss the likely impact of 'rising costs' for coffee shops on their profitability (Extract A, line 2).

(12)

(e) With reference to the information provided, discuss whether the coffee shop market is contestable.

(15)

(Total for question = 50 marks)

Q7.

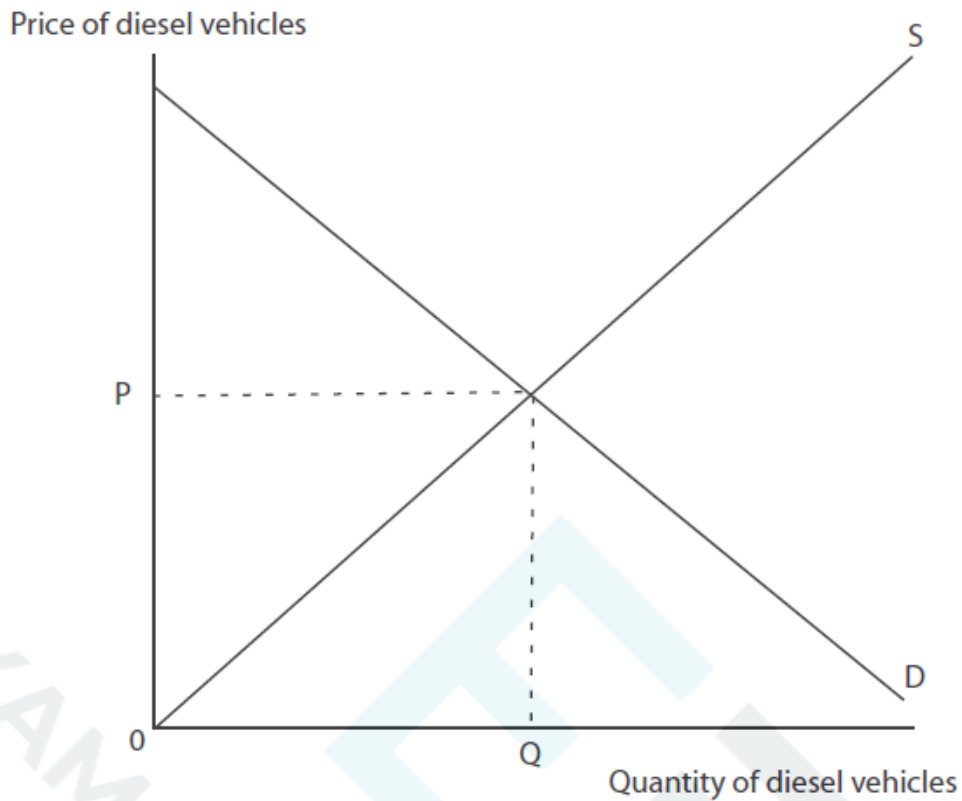
Use the data to support your answers where relevant. You may annotate and include diagrams in your answers.

The consensus that diesel emissions are bad for health has led to many cities planning to ban diesel vehicles from their centres. The mayors of Paris, Mexico City, Madrid and Athens have said they intend to ban these vehicles by 2025.

(Source: adapted from <http://www.autoexpress.co.uk/car-news/consumer-news/98747/diesel-ban-clean-air-zone-expansion-could-see-diesel-cars-charged-to>)

(a) Annotate the diagram to show the change in consumer surplus in the market for diesel vehicles, following the change in demand caused by the proposed 2025 ban in some city centres.

(2)



(b) A ban on diesel vehicles in city centres will reduce the demand for diesel fuel. This is because:

(1)

- ☐ **A** consumers are irrational
- ☐ **B** consumers expect the price of diesel to rise
- ☐ **C** the goods are complements
- ☐ **D** the goods are substitutes

(c) Define the term 'demand'.

(1)

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(Total for question = 4 marks)

Q8.

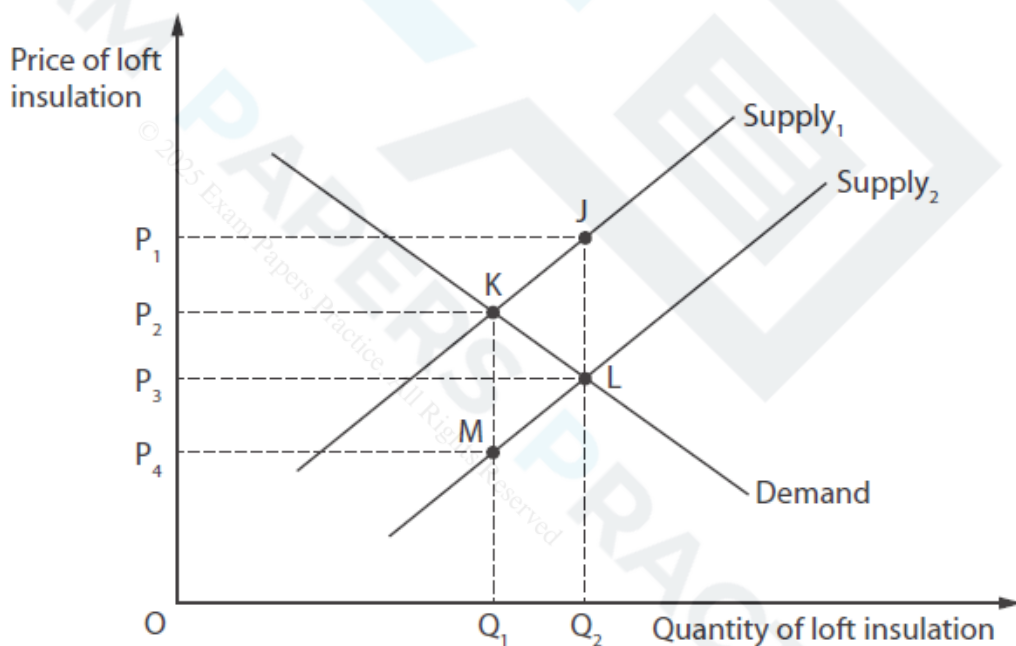
Use the data to support your answers where relevant. You may annotate and include diagrams in your answers.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☐ and then mark your new answer with a cross ☒.

The 'Energy Company Obligation' is the government's term for its programme to make houses in the UK more energy efficient.

(a) In the diagram below, the government grants a subsidy to energy suppliers to install loft insulation.

This shifts the market supply curve for loft insulation from Supply_1 to Supply_2 .



The total amount spent by the government on subsidies is represented by the area:

(1)

- ☒ **A** OP_1JQ_2
- ☒ **B** OP_3LQ_2
- ☒ **C** P_3P_1JL
- ☒ **D** P_4P_2KM

(b) Explain an opportunity cost to the government of granting subsidies to energy suppliers to install loft insulation.

(3)

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(Total for question = 4 marks)

Q9.

Read Figures 1 and 2 and the following extracts (A to C) before answering this question.

The travel and tourism industry

Figure 1: Package holiday market share of the six largest providers, booked by UK residents, 2019

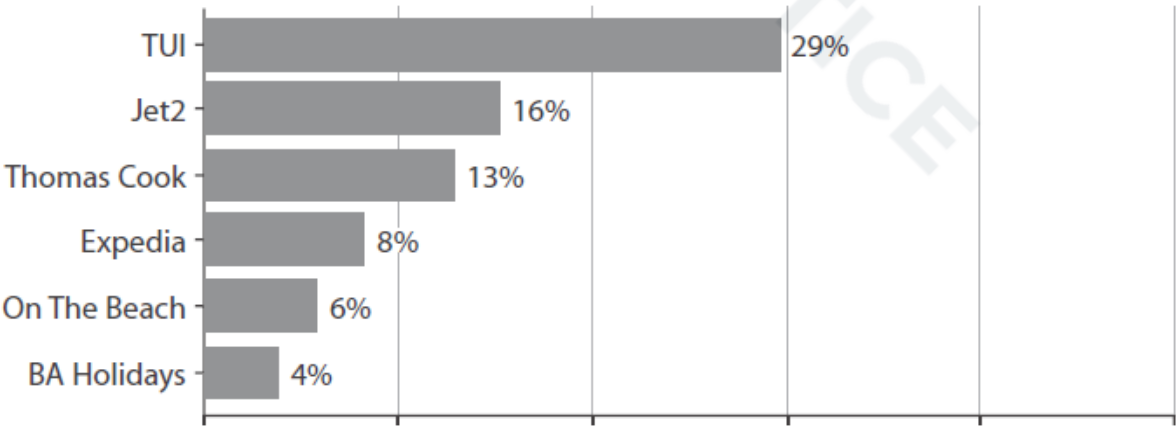


Figure 2: Jet2 package holiday prices to Sorrento

Price before Thomas Cook shutdown	Price 12 hours after Thomas Cook shutdown
2 Adults for 7 Nights from Monday 25 May 2020 🍴 Half Board 🛏 1 x Premium Double or Twin room ✈ Return flights from Edinburgh View flight times 🧳 22 kg Bag Allowance 🧳 10 kg Hand Baggage 🛡 ATOL Protected 🚗 Coach Transfers Holiday price from £1,576 Per person price £788	2 Adults for 7 Nights from Monday 25 May 2020 🍴 Half Board 🛏 1 x Premium Double or Twin room ✈ Return flights from Edinburgh View flight times 🧳 22 kg Bag Allowance 🧳 10 kg Hand Baggage 🛡 ATOL Protected 🚗 Coach Transfers Holiday price from £1,648 Per person price £824

Extract A

Marginal productivity of cabin crew

Cabin crew are responsible for loading passengers and providing in-flight meals. United Airlines is planning to reduce the number of its cabin crew members onboard international flights. The airline currently operates its planes with one more cabin crew member than its competitors. The marginal productivity of this additional crew member may be low. By reducing the number of its cabin crew members United Airlines will be able to operate more efficiently and compete more effectively.

5

Extract B

Thomas Cook's environmental impact

Thomas Cook Group plc's operations included its airline and 560 high street travel agents providing flights, hotels and package holidays.

The environmental impact of the travel industry is significant. It accounts for 8% of all global carbon emissions. Thomas Cook recognised the risks presented by climate change and actively engaged in reducing their airline emissions. Its plans included using more efficient aircraft and using lower-carbon fuel. In 2018, Thomas Cook was included in the top 10 of the world's most fuel-efficient airlines.

5

Extract C

Why did Thomas Cook shut down?

Thomas Cook Group plc ceased trading on 23 September 2019. The collapse of Thomas Cook left 600 000 travellers stranded overseas and approximately 21 000 worldwide employees were left without a job.

Thomas Cook's management said that the failure of rescue talks between banks, shareholders and the UK Government meant it had no choice but to shut down the business.

5

But in truth the tour operator's problems go back much further. A disastrous merger in 2007, increased debts, the internet revolution in holiday booking and Brexit uncertainty all contributed to the failure of the business.

10

In 2007 it merged with MyTravel. Thomas Cook directors had an objective of rapid company growth over short-term profitability. The merger was supposed to create a European giant, promising £75 million-a-year cost savings and a springboard to challenge emerging internet rivals. In reality, Thomas Cook was merging with a company that had only made a profit once in the previous six years, and the deal left the Group with huge debts. In May 2019, the firm reported a £1.5 billion loss.

15

The role of the management in Thomas Cook's collapse is being investigated by the UK Government. Thomas Cook executives' salaries and bonuses have been questioned. Directors received salaries totalling £20 million in the five years before its collapse. The Chief Executive Officer (CEO) earned a £500 000 cash bonus in 2017 and about £8.5 million in his five years with the company. It seems that around £4 million of this was in the form of shares. The share price reached £1.46 in 2018, but each share is now worthless.

20

The CEO said that the directors had worked "exhaustively" to rescue Thomas Cook and create a long-term turnaround strategy. "It is a matter of profound regret to me and the rest of the board that we were not successful."

25

The UK prime minister admitted that the government refused to grant £150 million as a subsidy to help rescue Thomas Cook in the short run. The UK prime minister stated: "Clearly, that is a lot of taxpayers' money and sets up, as people will appreciate, a moral hazard in the case of future such commercial difficulties that companies face. I have questions about whether it's right that the directors, or whoever, the board, should pay themselves large sums when businesses can go down the tubes like that. One is driven to reflect on whether the directors of these companies are properly incentivised to sort such matters out".

30

(a) Explain the likely impact of diminishing marginal productivity of labour on cabin crew staffing levels. Refer to Extract A in your answer.

(5)

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- (c) With reference to Extract C, assess whether Thomas Cook's failure was caused by the principal-agent problem.
- (10)
- (d) With reference to Extract C, discuss the proposed government subsidy to prevent Thomas Cook from reaching its shut-down point.
- (12)
- (e) With reference to the information provided, discuss the decision by Jet2 to increase its package holiday prices.
- (15)
- (Total for question = 50 marks)**

Q10.

(a) Define the term 'external benefits'.

(1)

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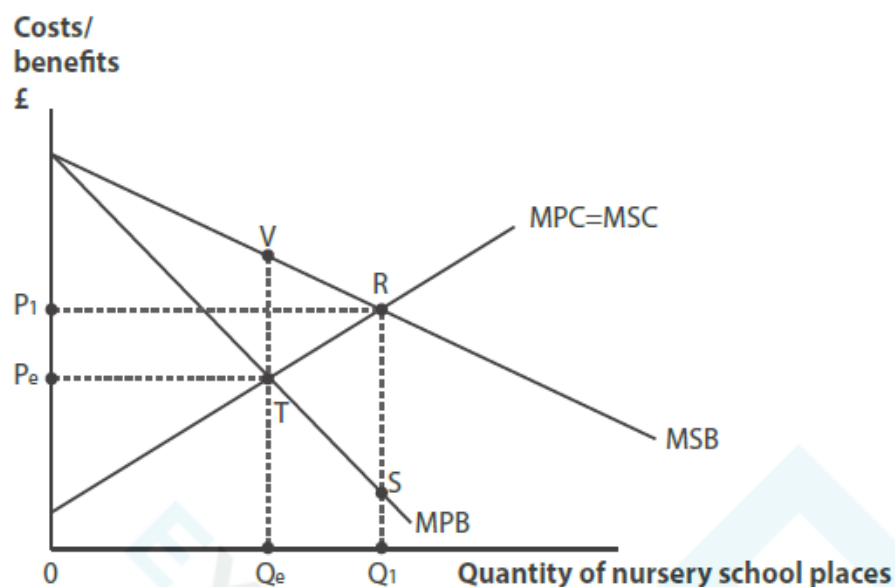
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Diagram A shows the market for nursery school places.

Diagram A



(b) Which **one** of the following measures could a government introduce to achieve the social optimum consumption of nursery school places?

It could introduce a:

- ☒ **A** maximum price of OP_1 per unit
- ☒ **B** subsidy of RS per unit
- ☒ **C** tax of VT per unit
- ☒ **D** minimum price of OP_1 per unit

(1)

(c) Annotate on Diagram A the welfare gain area and the new social optimum output resulting from successful government intervention in the market for nursery places.

(2)

(Total for question = 4 marks)